

R and R Studio

R Statistics

The statistical program **R** is nothing more than a programming language, mainly used for data manipulation and to perform statistical analyses. At the time of writing, this language is (one of) the leading program in statistics, although not the only programming language used by statisticians.

In order to use R, we need two things:

- a text editor in which to write our code
- a place to run this code

You can download R at <https://cran.r-project.org/mirrors.html>. Select the CRAN mirror site closest to you, then in the box labelled “Download and Install R”, click the link corresponding to your operating system.

R Studio

This is where RStudio comes handy. RStudio is an integrated development environment (IDE) for R. R and RStudio work together. R is a program that runs all your code, and RStudio is another program that allows you to control R in a more comfortable and friendly way. RStudio has the advantage of offering both a powerful text editor for writing your code and a place to run the code written in this editor. For these reasons, I highly recommend using RStudio instead of R. (Note that RStudio requires the prior installation of the R software provided by CRAN in order to be able to function properly. Just installing RStudio on your personal computer is not enough.)

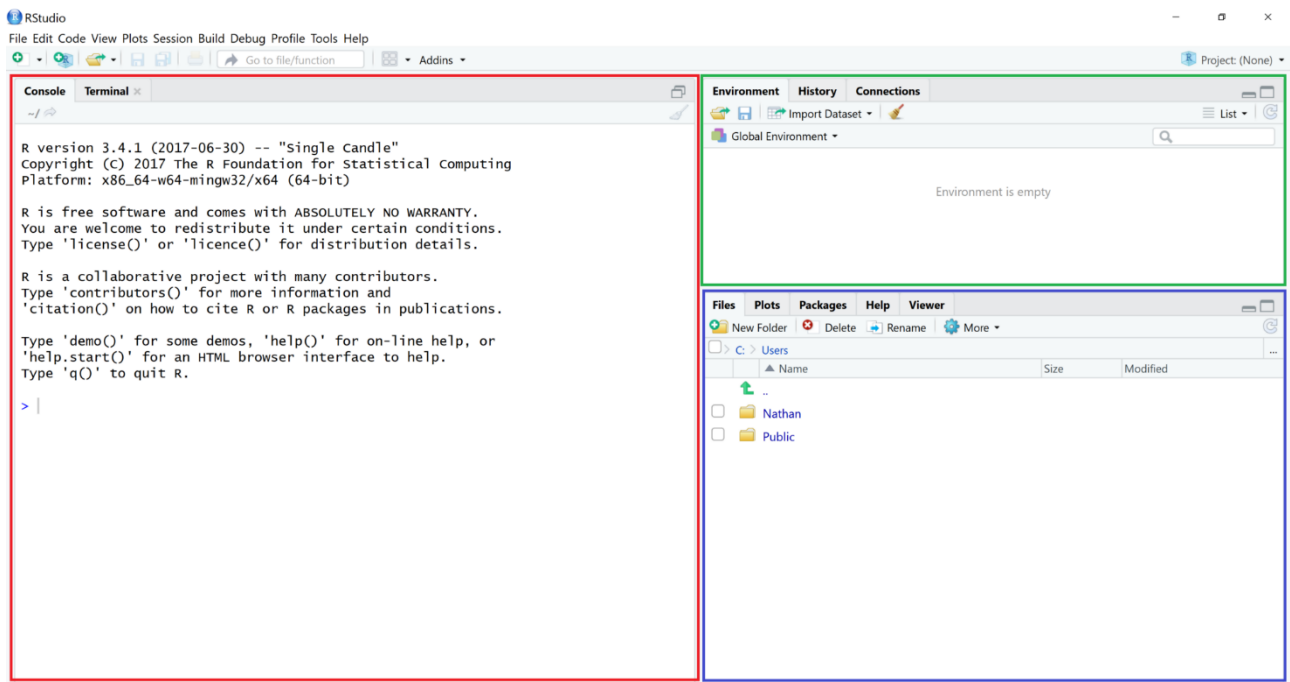
You can download RStudio at www.rstudio.com.

The main components of R Studio

Now that both programs are installed on your computer, let’s dive into the main components of RStudio.

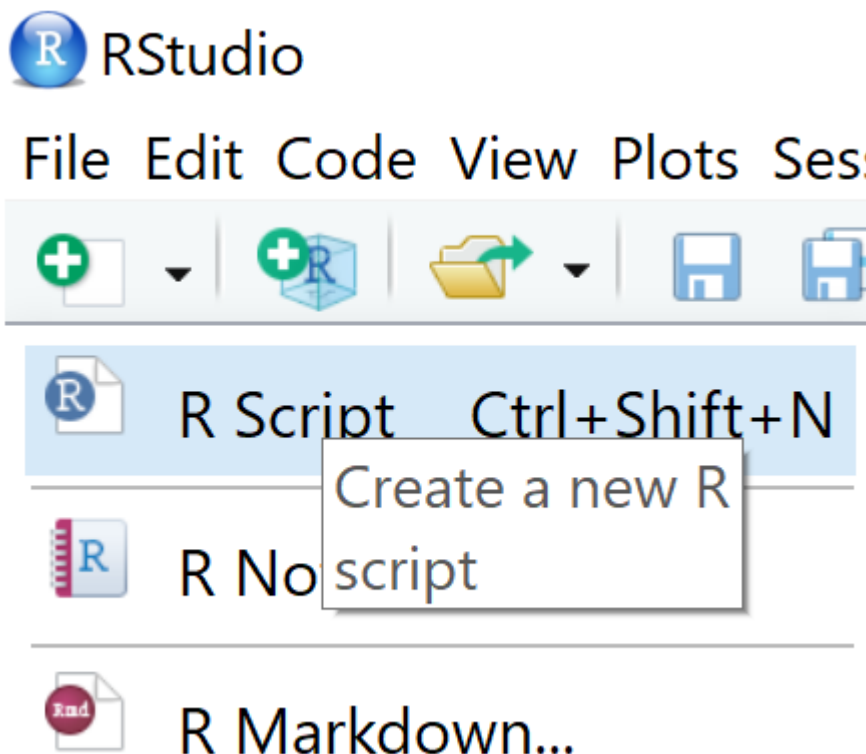
By default, the RStudio window has three panes:

1. The console (red pane)
2. The environment (green pane)
3. Files, plots, help, etc. (blue pane)



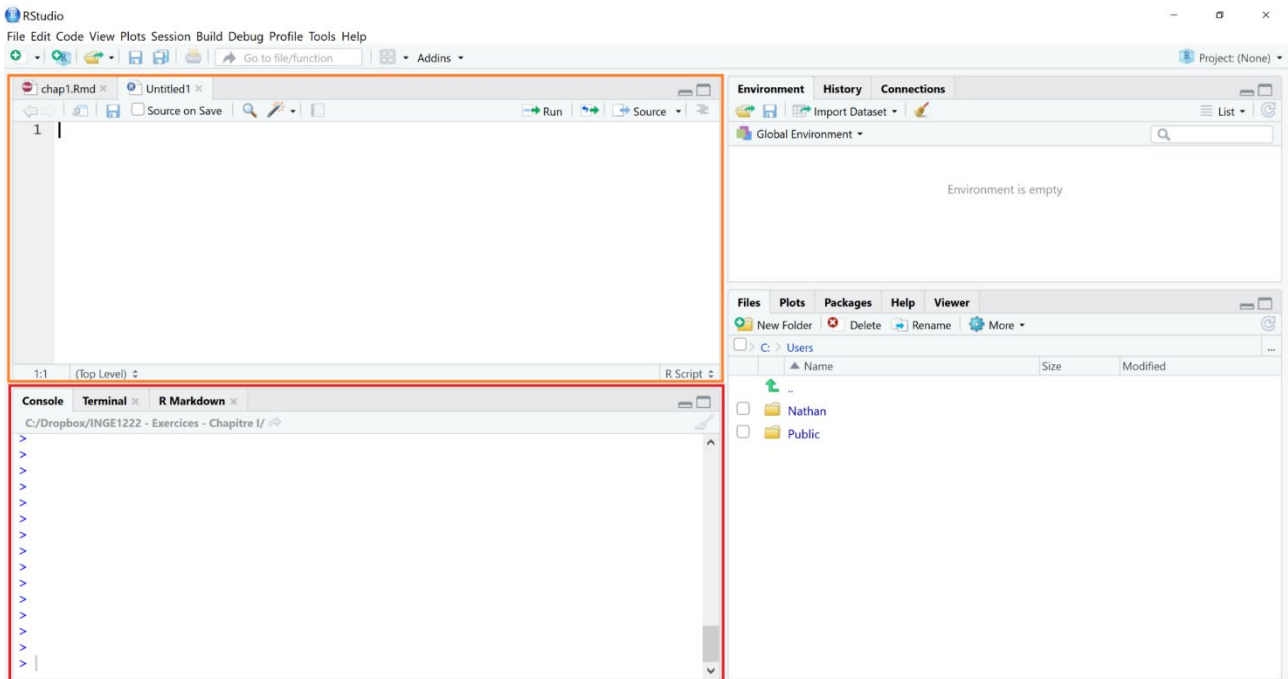
R Studio window

The console (red pane) is where you can execute your code (more information on the red and blue panes later). By default, the **text editor** does not open automatically. To open it, click on **File > New File > R Script** or click on the button representing a white sheet marked with a small green cross in the upper left corner, then on R Script:



New R script in R Studio

A new pane (in orange below), also known as the text editor, opens in which you will be able to write your code. The code will be executed, and the results displayed in the console (red pane).

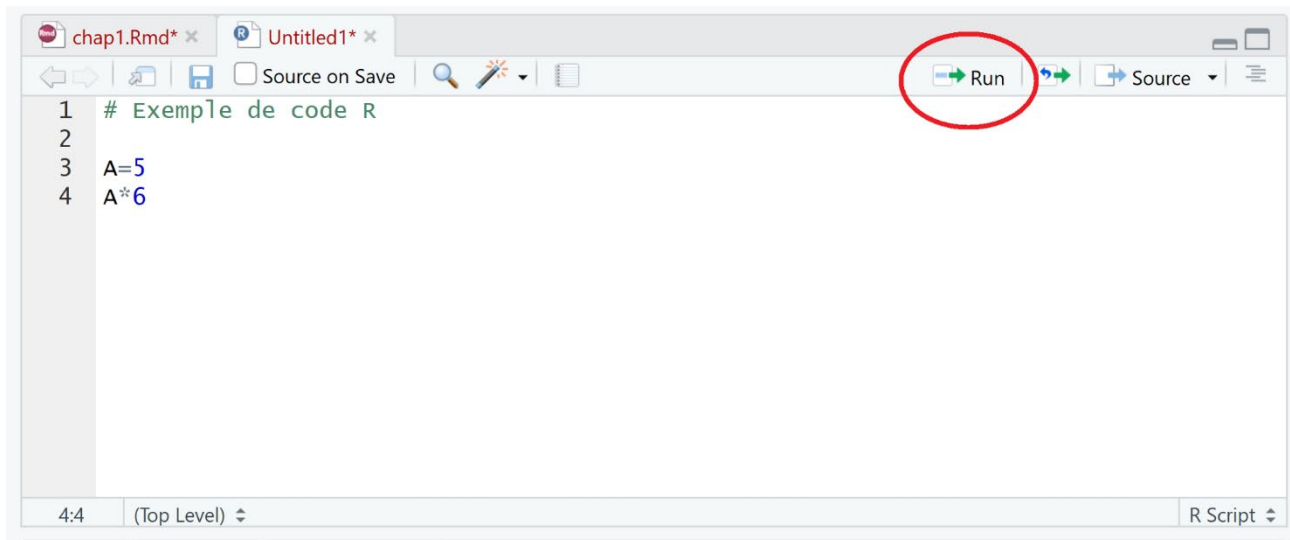


Write and execute your code in R Studio

Note that you can also write code in the console (red pane). However, I strongly recommend writing your code in the text editor (orange pane) because you can save the code written in the text editor (and thus execute it again later), while you cannot save the code written in the console.

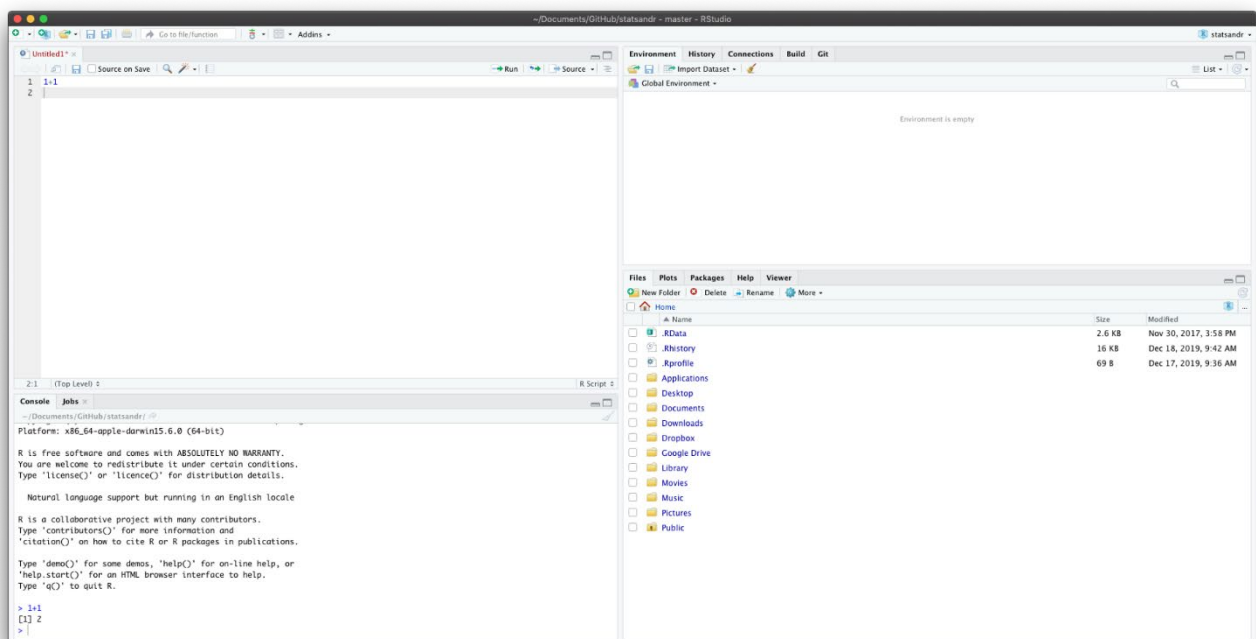
To execute code written in the text editor (orange pane), you have two options:

- Type your code and then press the “Run” button (see below) or use the keyboard shortcut CTRL + Enter (cmd + Enter on Mac). **Only the chunk** of code where your cursor is located will then be executed.
- Type your code and select in the text editor the part you want to execute and then press the “Run” button or use the keyboard shortcut CTRL + Enter (cmd + Enter on Mac). All the **selected** code will be executed



Run code in R Studio

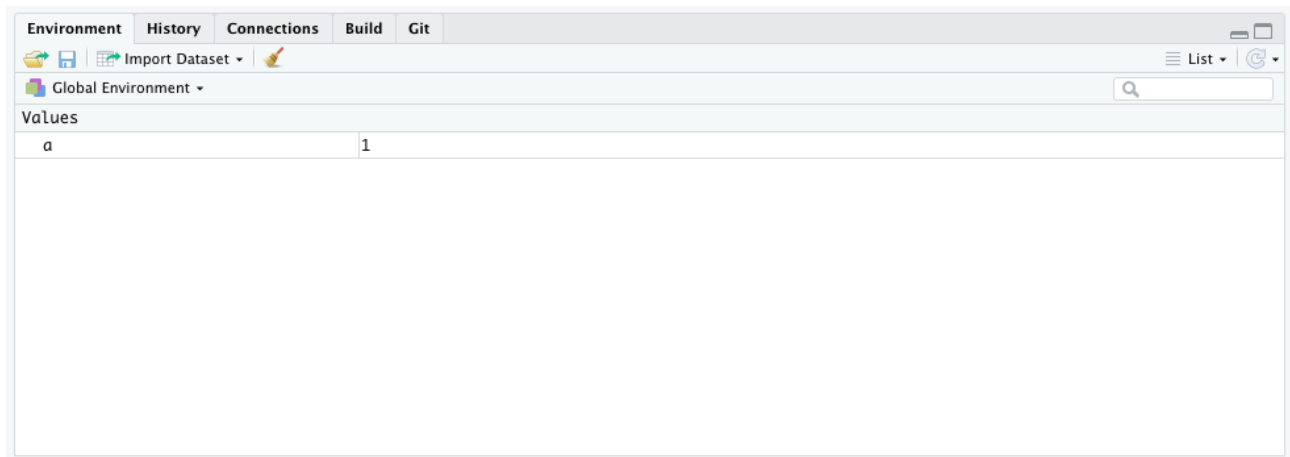
For example, try typing `1+1` in the text editor and execute it by clicking on “**Run**” (or CTRL/cmd + Enter). You should see the result `2` in the console, as shown in the screenshot below:



Example of code executed in RStudio

The text editor and the console are the panes you will use most often. The two other panes (the blue and green panes introduced earlier) will however still be very useful when using RStudio.

The environment (green pane) displays all values stored by RStudio. For example, if you type and execute the code `a = 1`, RStudio will store the value `1` for `a`, as shown in the screenshot below:



Environment pane in RStudio

This means that you can now perform any computations with `a`, such that if you execute `a + 1`, RStudio will render 2 in the console. In this pane you can also see a tab with a history of the code executed and a button to import a dataset (we will see more on importing a dataset in RStudio).

The last pane (blue) is where you will find everything else such as your files, the plots, the packages, the help documentation, etc. I discuss the Files tab in more detail [here](#) so let's discuss the other tabs:

- **Plot:** where you will see the rendered plots. For instance, run:

```
plot(1:10)
```

and you should see it in this tab. If you plotted more than one plots, you can navigate between them by clicking on the arrows. You can open the plot in a new window by clicking on Zoom and export your plot by clicking on Export. Those buttons are located just under the Plot tab (see figure below)

- **Packages:** where you see all your installed packages. Only fundamental functionalities come with R. Everything else must be installed from packages. Remind that R is open source; everyone can write code and publish it as a package. You are then able to use this package (and all functions built inside this package) for free. Some packages are installed by default, all others must be installed by running:

```
install.packages("name of the package")
```

 (do not forget `"` around the name of the package!).

Once the package is installed, you must load the package and only after it has been loaded you can use all the functions it contains.

To load a package, run:

```
library(name of the package)
```

Note. (this time `"` around the name of the package are optional, but can still be used if you wish).

You also have the possibility to install and load packages via the buttons under the Packages tab. For this, click on the button Install under Packages, type the name of the package you want to install and then click on Install. You will see that the code appears in the console. To load the package, find the package you want to load in the Packages window (you can use the search box), then click on the checkbox next to the name of the package. Again, the code is run in the console. See the figures below if needed. Note that you will need to install packages **only once**, but load packages **each**

time you open RStudio. Furthermore, note that an internet connection is required to install a package, while it is not required to load a package

- **Help:** documentation about all functions written for R. To access the help of a function, run `help("name of the function")` or simply `?name` of the function. For example, to see the help about the mean function, run:

`?mean.`

You can also press **F1** while having your cursor on a function